

Linear systems and elimination

Sharp row-deletion singular-value limit for spherical random matrices

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved

Independently reviewed resolution - 2026-09-11

Affirmative resolution. Theorem 1 and Sections 2-5 prove the displayed Gaussian trimmed-second-moment limit in probability along every sequence $n \rightarrow \infty$ and $m/n \rightarrow \infty$, with exactly $\lfloor \theta m \rfloor$ retained rows and the variational least singular value. Explicit failure-probability and error bounds are included; no faster aspect-ratio growth assumption is added.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects a sharp limit for the least singular value after adversarial row deletion; community impact includes random matrix theory and robust iterative solvers.

Problem statement

Fix $0 < \theta < 1$. For $A \in \mathbb{R}^{m \times n}$ define

$$s_\theta(A) = \min_{S \subseteq \{1, \dots, m\}, |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2 = 1} \|A_S x\|_2,$$

where A_S contains the rows indexed by S . Let $a_\theta > 0$ satisfy $\mathbb{P}(|G| \leq a_\theta) = \theta$ for $G \sim N(0, 1)$, and set

$$h_\theta = \frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2} dt.$$

For every integer sequence (m_j, n_j) with $n_j \rightarrow \infty$ and $m_j/n_j \rightarrow \infty$, let the rows of $A_j \in \mathbb{R}^{m_j \times n_j}$ be independent and uniformly distributed on $\mathbb{S}^{n_j-1}$. Is

$$\frac{s_\theta(A_j)^2}{\|A_j\|_2^2} \xrightarrow{\mathbb{P}} h_\theta?$$

This states Steinerberger's proposed asymptotic with $\theta = q - \beta$. Convergence in probability and the floor convention make the source's limiting statement explicit. The source also asks for quantitative error bounds; these belong to this problem rather than separate entries.

Connection to numerical linear algebra

The quantity measures the worst conditioning remaining after rows are discarded. It controls convergence guarantees for quantile Kaczmarz methods on corrupted linear systems.

References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, equation (8) and the following paragraph. [Published paper](#). [Author preprint](#), v1 (2021), §2.3, equation ($\diamond$).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1, discussion of Steinerberger’s random-matrix heuristic and the distinction between static and streaming data; §5. [Preprint](#).

Earlier status check — 2026-09-08

Checked the published formulation and the arXiv version history (2107.05554 has only v1), then searched the problem’s title, the author’s name with “subsingular”, and combinations of “trimmed/smallest singular value”, “quantile”, “Kaczmarz”, “heuristic”, “proof”, “2025”, and “2026”. The August 2026 streaming paper still describes the original estimate as a heuristic; its fresh-sample convergence bounds do not establish this static-matrix limit. No later proof or counterexample was located in this bounded search. That is a literature-status assessment, not a proof of openness.

Audit update — 2026-09-10

Rechecked Steinerberger’s [primary formulation](#), §2.3, and the [2026 streaming follow-up](#). Searches found no proof of the static proportional-deletion limit. Cai, Chen, Ma, and Wu’s new [subsample-size theorem](#), SIAM J. Matrix Anal. Appl. **47** (2026), 802–823, concerns quantile estimation during QRK, a different asymptotic quantity.